1.10 Early Finishers: Sequences and Compound Interest — Markscheme

Extension set, scored separately from the quiz. Calculator allowed. Total: 19 marks. Answers are exact or correct to three significant figures.

1. [Maximum mark: 11]

   $u_3 + u_8 = 48,\ \ u_{12} = 5\,u_2.$

   (a) Write each term as $u_1 + (n-1)d$:
   $(u_1 + 2d) + (u_1 + 7d) = 48$ **M1**
   $2u_1 + 9d = 48$
   $u_1 + 11d = 5(u_1 + d)$ **M1**
   $4u_1 = 6d$
   $u_1 = 1.5d$
   $2(1.5d) + 9d = 48$
   $12d = 48$
   $d = 4$ **A1**
   $u_1 = 1.5(4) = 6$ **A1**

   One M1 for each condition written in terms of u_1 and d. Accept solving the system $2u_1 + 9d = 48$, $4u_1 - 6d = 0$ on the GDC. FT: award the final A1 for u_1 correct from the candidate's own d.

   (b) $6 + 4(n-1) < 500$ **M1**
   $4(n-1) < 494$
   $n - 1 < 123.5$
   $n < 124.5$ **A1**
   124 terms **A1**

   Accept an equation in place of the inequality, or a table search showing $u_{124} = 498$ and $u_{125} = 502$ (markscheme-spec §18). The final A1 is for the count, 124, not for 124.5. FT from the candidate's u_1 and d.

   (c) The 1st, 2nd and 4th terms are $3,\ \ 3 + d,\ \ 3 + 3d.$ **(M1)**
   $\dfrac{3+d}{3} = \dfrac{3+3d}{3+d}$ **M1**
   $(3+d)^2 = 3(3+3d)$
   $d^2 - 3d = 0$
   $d \neq 0$, so $d = 3$ **A1**

   The terms are $3,\ 6,\ 12$, so $r = 2$ **A1**

   Second M1 for equal ratios, or the equivalent $(3+d)^2 = 3(3+3d)$. Award A0 for $d = 3$ if $d = 0$ is given as a second answer and not rejected. FT: award the final A1 for r correct from the candidate's own d. $d = 0$ also solves the equation (the constant sequence $3, 3, 3$), which is why the question excludes it.

2. [Maximum mark: 8]

Sofia: $PV = 5000$, 4.8% compounded monthly. Luca: $PV = 6000$, r% compounded quarterly, $FV = 9000$ after 10 years.

(a) $FV = PV\left(1 + \frac{r}{100k}\right)^{kn}$, with $k = 12$:

$5000\left(1 + \frac{4.8}{100(12)}\right)^{12n} > 10\,000$ **M1**

n (years)	Value
14	\$9777.63
15	\$10 257.42

A1

15 years **A1**

M1 for the substituted expression, or the TVM entries $I\% = 4.8$, $PV = -5000$, $PMT = 0$, $FV = 10\,000$, $P/Y = C/Y = 12$. First A1 for evidence that 14 years is not enough: the 14-year value, or $N = 173.6$ months $= 14.47$ years. A bare "15" with no working earns M0A0A1.

(b) $6000\left(1 + \frac{r}{100(4)}\right)^{4(10)} = 9000$ **M1**

$\left(1 + \frac{r}{400}\right)^{40} = 1.5$

$r = 400\left(1.5^{1/40} - 1\right)$

$r = 4.08 \quad (4.07527\ldots)$ **A1**

A1 for 4.08 (3 s.f.) or the calculator display with an ellipsis (4.07527…); 4.075 or 4.0753 with no ellipsis earns A0. Accept the TVM solver: $N = 40$, $PV = -6000$, $PMT = 0$, $FV = 9000$, $P/Y = C/Y = 4$, solve for I%.

(c) Sofia: $S_n = 5000\left(1 + \frac{4.8}{100(12)}\right)^{12n}$ Luca: $L_n = 6000\left(1 + \frac{4.0752\ldots}{100(4)}\right)^{4n}$ **M1**

n (years)	Sofia	Luca
24	\$15 786.26	\$15 877.07
25	\$16 560.90	\$16 534.06

A1

Year 25 **A1**

M1 for both expressions, or a GDC table of both. First A1 for the adjacent-year comparison: Luca ahead at year 24, Sofia ahead at year 25. Accept values from $r = 4.08$ rounded (Luca \$15 894.92 and \$16 553.42); the answer is still year 25. FT from the candidate's r in (b). Sofia starts with less but earns the higher effective rate (4.91% a year against 4.14%), so she catches up slowly.